



ISO 14971:2019 +A11:2021

Risk Analysis Workshop

Part B: Risk Analysis techniques + Hands-on exercise

Training Session | Nordberg Medical | Orderly People

Session Overview

~40 min

Phase 1: Examples and Techniques

Risk analysis examples + Techniques to support risk analysis

~20 min

Phase 2: Hands-on Exercise (Breakout rooms)

Complete the risk analysis matrix.

~20 min

Phase 3: Review & Discussion

Present results, discuss challenges, and review learnings.

Quick Recap: Risk Assessment and Control



Key Reminders



Hazard ≠ Harm. A hazard is a potential source of harm. The harm is the actual injury or damage (to people, property, or environment)



Risk control measures (RCM): Always consider the 3-tier hierarchy (Safety by design; Protective measures; Information for safety)



Verification of Risk Control Measures: **Implementation** (did you DO it?), is different from **effectiveness** (does it WORK?)

What it can look like in practice →

Risk Analysis Matrix (1/2)

Continues on next slide →

Worked example: PLLA dermal filler, Initial risk assessment + Risk control measures

Risk ID	Hazard	Sequence of Events	Hazardous Situation	Harm	P	S	Initial Risk ACC ?	Risk Control Measures
R-001	Mechanical hazard	Practitioner uses incorrect injection technique (too superficial into the dermis instead of deep dermis / subcutaneous layer).	PLLA microspheres deposited in superficial dermis	Visible nodule formation	P3	S2	(ACC)	1. Depth marking on injection needle 2. Training: recommended technique with visual depth guidance 3. IFU specifies injection depth (deep dermis / subcutaneous)
R-002	Mechanical hazard	Intravascular injection during nasolabial fold treatment. Needle penetrates blood vessel; PLLA microspheres enter vasculature.	PLLA microspheres present in blood vessel	Vascular occlusion	P2	S4	NOT ACC	1. IFU: aspiration guidance before each injection pass 2. IFU: vascular occlusion warning and emergency protocol 3. Training: facial vascular anatomy awareness
R-003	Use error	User confuses reconstitution volumes for DF001 and DF002. Wrong volume used, resulting in incorrect concentration.	Suspension at wrong concentration injected into patient	Adverse tissue reaction	P3	S3	NOT ACC	1. Color-coded labels; blue = DF001, green = DF002 2. Design: differentiated vial sizes 3. IFU: prominent reconstitution volume table with visual icons
R-004	Biological hazard	Failure to follow aseptic technique during reconstitution. Bioburden introduced.	Contaminated PLLA suspension injected into patient tissue	Injection-site infection	P3	S3	NOT ACC	1. IFU: aseptic reconstitution technique instructions 2. Design: single-use vial with narrow-neck opening 3. Training: reconstitution procedure step-by-step video
R-005	Chemical hazard	Reconstituted product used beyond the 72-hour post-reconstitution stability window. Product degrades; CMC Na carrier loses viscosity.	Degraded / unstable suspension injected into patient	Reduced efficacy	P3	S2	(ACC)	1. Label: 72 h use-by label printed on vial 2. IFU: discard instructions and stability window warning

Risk Analysis Matrix (2/2)

Continued from previous slide, now adding: Verifications of RCM, Residual risk, Benefit/Risk references, and New risks arising from implemented risk controls.

Risk information (summary of previous slide)	Risk Control Measures (from previous slide)	Verification of Implementation of RCM	Verification of Effectiveness of RCM	rP	rS	Residual Risk ACC?	If NOT ACC: Benefit-Risk Analysis ref.	New Risks Arising from RCM?
R-001: Mechanical Incorrect injection technique, too superficial Harm: Nodule	1. Depth marking on injection needle 2. Training: recommended technique with visual depth guidance 3. IFU specifies injection depth (deep dermis / subcutaneous)	1. Needle marking per spec (DVR-045 v1) 2. Training material approved (TRN-008 v2) 3. IFU content review completed (DOC-102 v3)	1, 3. Simulated-use usability study with 15 practitioners (Report RPT-061) 2. Training effectiveness records All: Post-market complaint rate monitoring (PMS-004)	P2	S2	ACC	N/A	Depth marking may be confused with other needle markings. → Assess as R-006
R-002: Mechanical Intravascular injection, into blood vessel Harm: Vascular occlusion	1. IFU: aspiration guidance before each injection pass 2. IFU: vascular occlusion warning and emergency protocol 3. Training: facial vascular anatomy awareness module	1. IFU content review completed (DOC-102 v3) 2. Emergency protocol included (DOC-115 v1) 3. Training module approved (TRN-011 v1)	1. IFU Usability study; and Literature review on aspiration test efficacy (LIT-023) 2. PMCF study monitoring vascular events (CIP-003, ongoing) 3. Training effectiveness records	P1	S4	(ACC)	N/A	Aspiration step may delay procedure and cause additional patient discomfort. → Assess as R-007
R-003: Use error Confuses DF001 for DF002, Wrong concentration injected Harm: Tissue reaction	1. Color-coded labels; blue = DF001, green = DF002 2. Design: differentiated vial sizes 3. IFU: prominent reconstitution volume table with visual icons	1. Label color specs (DVR-051 v1) 2. Vial design specs (DVR-052 v1) 3. IFU review completed (DOC-102 v3)	All: Simulated-use usability study with both variants available (Report RPT-067)	P1	S3	ACC	N/A	N/A
R-004: Biological Non-aseptic reconstitution technique, bioburden Harm: Infection	1. IFU: aseptic reconstitution technique instructions 2. Design: single-use vial with narrow-neck opening 3. Training: reconstitution procedure step-by-step video	1. IFU content reviewed (DOC-102 v3) 2. Vial neck design spec (DVR-048 v1) 3. Training video approved (TRN-009 v1)	All: Simulated reconstitution study with bioburden sampling (Report RPT-058)	P3	S3	NOT ACC	Benefit-risk record ref. BRA-004 v1: Residual risk acceptable when compared to SOTA & similar devices.	N/A
R-005: Chemical Reconstituted product used after 72-hour window Harm: Reduced efficacy	1. Label: 72h use-by label printed on vial 2. IFU: discard instructions and stability warning (72h)	1. Label spec (DVR-055 v1) 2. IFU review completed (DOC-102 v3)	All: Usability studies on label and IFU (Report RPT-070)	P1	S2	ACC	N/A	N/A

NOT ACC risks (R-004) require documented Benefit-Risk Analysis. New risks identified (R-006 to R-007) must re-enter the risk management process from ISO14971 §5 (Risk analysis).

ISO/TR 24971 Annex B

Risk Analysis techniques

Methods to support risk analysis

Risk analysis techniques

ISO/TR 24971 dives deeper into various techniques to support risk analysis, including the following:

Early development stage

- PHA – Preliminary Hazard Analysis
- FTA – Fault Tree Analysis
- ETA – Event Tree Analysis
- HAZOP – Hazard and Operability Study

Late development stage

- FMEA – Failure Mode and Effects Analysis (FMEA)
- HACCP – Hazard Analysis and Critical Control Point

Preliminary Hazard Analysis (PHA)

ISO/TR 24971:2020, Annex B.2 | Best for: Early design phase

What is it?

The traditional ISO14971 risk analysis workflow;

- identify hazards,
- events,
- hazardous situations, and
- potential harms.

When and why use it?

When: At the very **START** of development, when detailed design information is limited, and prior to establishing Design Input requirements (Risk Controls).

Why: To establish a baseline of all known and foreseeable hazards before investing in detailed design. PHA outputs feed directly into your ISO 14971 risk analysis and guide where to focus engineering effort.

How to use it?

1. Assemble a cross-functional team (engineering, quality, clinical, regulatory)
2. Define intended use and reasonably foreseeable misuse
3. Identify hazards - Considering materials, components, interfaces, operating principles, and use environment, etc.
4. For each identified hazard, describe sequence of events, hazardous situation, and harm
5. Estimate initial risk (P and S) and identify potential risk controls
6. Record in a tabular format - update iteratively as design matures

Example

Refer to next slide →

PHA Example: Implantable Cardiac Pacemaker

Preliminary Hazard Analysis - identifying hazards, hazardous situations, and harms in early design phase

Hazard	Sequence of Events	Hazardous Situation	Harm	P	S	Initial Risk Control Ideas
Biocompatibility	Lead insulation material degrades over time; toxic compounds leach into surrounding tissue	Patient's cardiac tissue is exposed to toxic degradation products from the device lead	Chronic inflammation	P3	S4	Biocompatible material selection per ISO 10993; accelerated ageing testing
Electromagnetic hazard	Patient enters MRI environment; pacemaker antenna effect causes lead tip heating	Excessive thermal energy is delivered to the myocardium (heart muscle) at the lead tip	Myocardial damage	P3	S5	MR-conditional design; filtered lead; IFU warnings about MRI exposure
Electrical hazard	Battery approaches end-of-life; voltage drops below minimum pacing threshold	Pacemaker-dependent patient does not receive pacing stimulus when needed	Cardiac arrest	P2	S5	Elective Replacement Indicator (ERI); remote monitoring alert; IFU replacement schedule

Key characteristics of a PHA table: The PHA captures the complete chain from hazard to harm, with initial risk estimates and early control ideas. It is deliberately high-level - you refine it as the design matures. The team should include clinical, engineering, quality, and regulatory perspectives. Each row represents one hazard-to-harm pathway. A single hazard can appear in multiple rows with different sequences leading to different harms.

Fault Tree Analysis (FTA) IEC 61025

ISO/TR 24971:2020, Annex B.3 | Best for: Finding causes of a known undesired consequence/harm (or hazardous situation)

What is it?

A deductive (top-down) technique that starts with a known undesired consequence - the 'top event' (e.g. a specific harm or hazardous situation) - and works backwards to find all possible combinations of causes.

The result is a tree-shaped diagram using logic gates (AND / OR) showing how lower-level events combine to cause the top event.

How to use it?

1. Define the **top event** precisely (e.g. 'patient receives overdose')
2. Ask: 'What could cause this?' - identify immediate causes
3. Connect causes with logic gates: **OR** (any single cause sufficient) or **AND** (multiple causes required together)
4. Continue decomposing each cause until you reach basic events (component failures, human errors, environmental conditions)
6. Identify critical paths for risk control focus

When and why use it?

When: Use when you need to deeply investigate a specific critical consequence - particularly one with high severity (e.g., patient death, blindness).

Why: Finding all causes of a single selected harm / undesired consequence.
FTA can illustrate when two or more events must occur together to create the consequence. Also great for PMS investigations (root cause analysis).

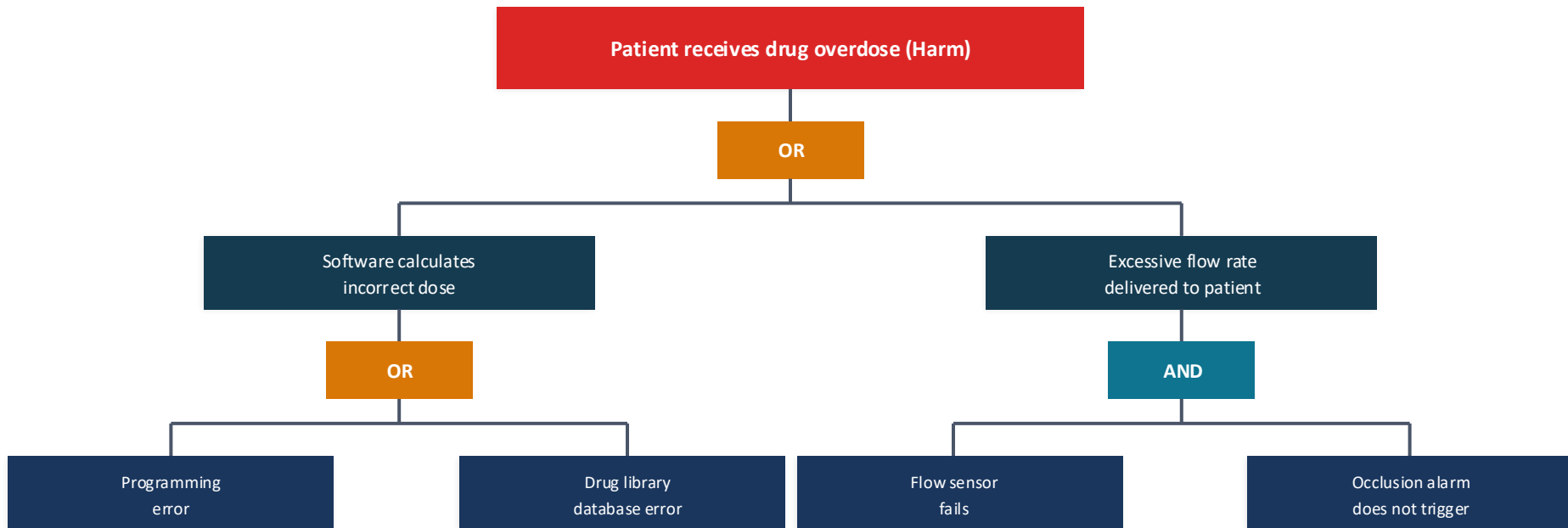
Example

Top event: 'Patient receives **drug overdose** from infusion pump.'

The tree reveals this requires EITHER a software dosing error OR a flow sensor failure AND a missing alarm. The AND gate shows two independent safeguards must both fail - this insight helps justify the residual risk level and guides testing priorities.

FTA Example: Infusion Pump Drug Overdose

Fault Tree Analysis - working backwards from a top event through AND/OR logic gates to find root causes



OR

Any single cause is sufficient

AND

All causes must occur together

Key insight: The AND gate on the right branch means BOTH the flow sensor AND the alarm must fail for an overdose via that path. This is why redundant safeguards reduce risk - the probability is $P(\text{sensor}) \times P(\text{alarm})$, which is much lower than either alone. The OR gate means any single software error path is sufficient to cause the overdose - a higher-priority area for risk control.

Event Tree Analysis (ETA) IEC 62502

ISO/TR 24971:2020, Annex B.4 | Best for: Tracing various outcomes of an initiating event (and calculating their probabilities)

What is it?

An inductive (bottom-up) technique that starts with an **initiating event** (e.g., a device malfunction or use error) and traces all possible outcomes forward through a branching tree.

When and why use it?

When: Use when you want to understand ALL the possible consequences of a specific triggering event, including how existing safeguards change the outcome.

Why: ETA maps the '**what happens next**' chain. It is the natural complement to FTA: FTA answers 'what causes this event?', ETA answers 'what happens after this event?' Together they provide a complete picture.

How to use it?

1. Identify the **initiating event** (device malfunction, use error, external event)
2. List the subsequent events (e.g., safeguards) in chronological order
3. For each event, branch into 'works' (top) and 'fails' (bottom)
4. Trace each path to its end outcome
5. Assign probabilities to each branch if data is available
6. Calculate end-state probabilities by multiplying along each path
7. Identify which events/safeguard failures lead to the most severe outcomes

Example

Initiating event: 'Ventilator pressure sensor fails' (possible lung damage)

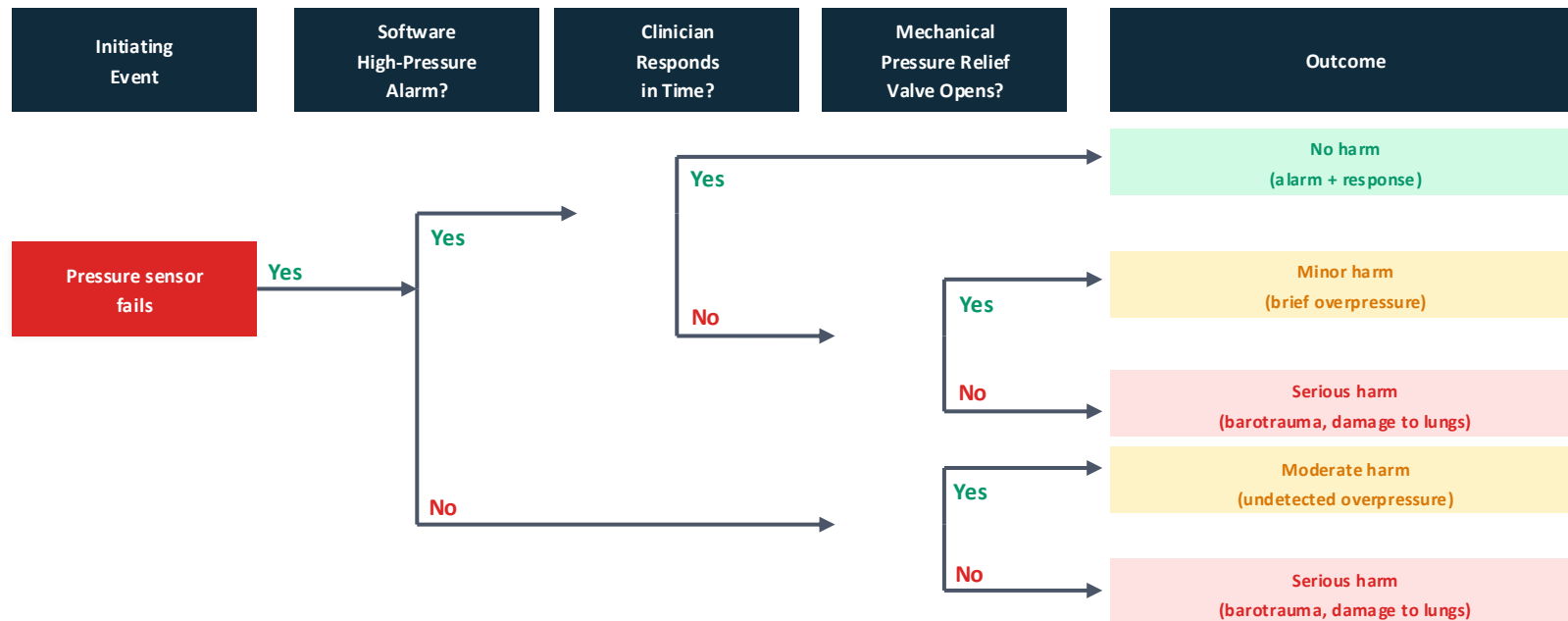
Safeguards:

- 1) Software alarm triggers?
- 2) Clinician responds?
- 3) Backup pressure relief valve opens?

Tracing all yes/no combinations gives various outcomes. This shows which safeguards are most critical to maintain.

ETA Example: Ventilator Pressure Sensor Failure

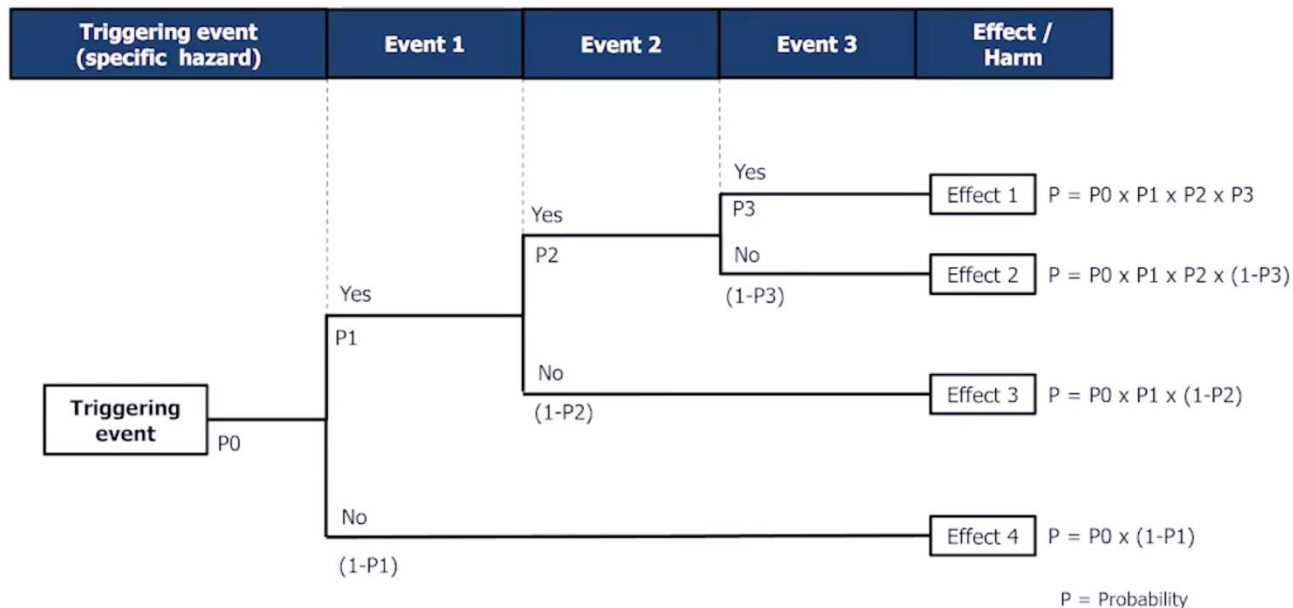
Event Tree Analysis - tracing forward from an initiating event through events (e.g., safety barriers) to all possible outcomes



Reading the tree: Follow each path from left to right. Every 'Yes' (barrier works) moves the outcome towards safety; every 'No' (barrier fails) moves towards harm. The worst case (bottom right) requires ALL three barriers to fail simultaneously. If you can show each barrier has independent reliability, the combined probability of the worst case is extremely low. ETA reveals which barriers contribute most to safety and where redundancy is valuable.

ETA Example, Probability of various harms

Event Tree Analysis – Useful for finding probabilities of various harms



Failure Mode and Effects Analysis (FMEA) IEC 60812

ISO/TR 24971:2020, Annex B.5 | *Best for: Systematic component/process failure analysis. How can this component fail, and what are the effects?*

What is it?

A system **reliability** technique that examines **components or process steps** and asks: 'How could this fail, and what are the effects?'

For each failure mode, the causes, effects, and severity are documented.

- Design FMEA (dFMEA) for product design
- Process FMEA (pFMEA), e.g., for manufacturing process or other QMS processes.

When and why use it?

When: Use during mature development phase when you have **enough design detail to analyse individual components (or process steps)**.

Why: FMEA is excellent at catching single-fault conditions systematically. However, FMEA alone is NOT sufficient for ISO 14971 compliance - it analyses fault conditions only, not normal-use hazards.

How to use it?

1. Define the scope: Design FMEA (components) or Process FMEA (manufacturing steps)
2. For each item, list all potential failure modes
3. For each failure mode, identify: cause(s), effect(s), Severity (S)
4. Estimate Probability of occurrence (P) and Detectability (D)
5. Estimate risks by established criteria, or $RPN = S \cdot P \cdot D$ (temporarily for FMEA)
6. Define risk control measures
7. Transfer findings to ISO14971 risk analysis records

Example

pFMEA for **catheter tip bonding**:

- Failure mode = 'incomplete bond.'
- Cause = insufficient dwell time in heat station.
- Effect = tip separation during use.
- Severity = critical (vessel perforation).
- Controls: validated bonding parameters + 100% pull-test.

The pFMEA ensures every manufacturing step has been challenged for failure.

FMEA Example: Design FMEA for Catheter Assembly

Failure Mode and Effects Analysis - systematically examining each component for failure modes, causes, and effects

Component	Function	Failure Mode	Cause of Failure	Effect of Failure	S	P	Risk Acc?	Risk Control
Balloon	Dilate vessel lumen to target diameter	Balloon rupture during inflation	Material defect; over-inflation beyond burst pressure	Vessel wall injury; retained fragments	S4	P2	NOT ACC	Burst pressure testing per lot; max inflation pressure in IFU; material incoming inspection
Tip bond	Secure distal tip to catheter shaft	Incomplete bond (tip separation)	Insufficient heat dwell time in bonding process	Tip embolisation; vessel occlusion	S5	P2	NOT ACC	Validated bonding parameters; 100% pull-test; X-ray inspection
Guidewire lumen	Allow guidewire passage for navigation	Lumen kink or collapse	Excessive bending beyond min. radius during procedure	Inability to advance or withdraw guidewire	S3	P3	NOT ACC	Kink resistance testing; braided shaft reinforcement; minimum bend radius in IFU

FMEA vs ISO 14971 risk analysis: Notice the FMEA focuses on component failures. These are fault conditions only. A device can work perfectly and still pose hazards during normal use (e.g. vessel perforation during normal catheter navigation). That is why FMEA alone does not satisfy ISO 14971 - you need PHA or equivalent to capture normal-use hazards. Use FMEA as a complementary tool feeding into your overall risk analysis.

Hazard and Operability Study (HAZOP) IEC 61882

ISO/TR 24971:2020, Annex B.6 | *Best for: multi-step processes with defined design intent. Can also be applied to the operation of a medical device.*

What is it?

A structured technique that systematically identifies deviations from design intent using 'guide words' applied to process parameters. HAZOP assumes hazardous situations are caused by **design deviations or operational variations**.

Guide words (More, Less, Early, Late, etc) are combined with **parameters** (flow, temperature, concentration, time, etc) to generate '**what if**'-scenarios.

Each credible deviation is then assessed for consequences.

When and why use it?

When: To analyze multi-step processes where each step has a defined design intent (e.g., manufacturing), or to analyze e.g., methods used for the diagnosis or treatment of disease. Can be performed early in development, but **requires established design requirements (design intent)**.

Why: Forces **structured 'what if' thinking** that catches deviations that might not otherwise be considered. It is particularly strong at identifying operability problems and use errors that arise from process parameter variations.

How to use it?

1. Define the nodes to analyze (distinct process steps or device functions)
2. For each node, identify the design intent and key parameters
3. Apply guide words to each parameter (e.g. 'No flow', 'More temperature')
4. For each credible deviation: identify causes, consequences, and existing safeguards
5. Assess risk level and identify risk controls
6. Transfer relevant findings to ISO14971 risk analysis records

Example - HAZOP on a sterile filling process

Node (a distinct step in a process): "Transfer solution to vials"

Parameter: Temperature.

Guide word: 'More.'

Deviation: 'Solution temperature too high during fill.'

Consequence: Protein degradation

Cause: Heat exchanger malfunction.

Control: Inline temperature sensor with automatic fill-stop.

HAZOP Example: Sterile Filling Process

Hazard and Operability Study - applying guide words to process parameters to find deviations from design intent

Node (distinct process step): Transfer sterile solution from bulk tank to vials → Design Intent: Deliver 5.0 mL ± 0.1 mL per vial at 20°C ± 2°C

Parameter	Guide Word	Deviation	Possible Cause	Consequence	Existing Safeguard	Action Required?
Volume	More	Fill volume > 5.1 mL	Pump calibration drift; valve sticking open	Overfill: product waste; potential vial overflow contamination	In-line volume check; reject station	Validate check weight limits
Volume	Less	Fill volume < 4.9 mL	Pump wear; air bubble in line; partial blockage	Underfill: sub-therapeutic dose to patient	In-line volume check; reject station	Add bubble detector
Volume	No	No fill delivered	Pump failure; upstream valve closed; tank empty	Empty vial reaches patient; no treatment	Weight check rejects empty vials	Add tank level alarm
Temperature	More	Solution > 22°C	Heat exchanger fault; ambient temperature rise	Material degradation	In-line temperature sensor	Auto-stop if temperature exceeded
Temperature	Less	Solution < 18°C	Chiller over-cooling; winter conditions	Viscosity change; fill accuracy affected	In-line temperature sensor	Low temp. alarm

The guide word system: By systematically combining each **parameter** (volume, temperature, time, pressure, etc) with each **guide word** (No, More, Less, Reverse, Other Than, Early, Late), the team ensures no deviation is overlooked. The highlighted guide words force structured thinking.

Hazard Analysis & Critical Control Points (HACCP)

ISO/TR 24971:2020, Annex B.7 | Systematic, process-focused | *Best for: critical process control*

What is it?

A 7-step systematic approach that identifies hazards in a process, determines Critical Control Points (CCPs) where control is essential, and establishes monitoring systems to ensure ongoing safety.

Originally developed for food safety, HACCP focuses on prevention through process control rather than reliance on end-product testing alone.

When and why use it?

When: Use for manufacturing processes where specific steps are critical to product safety and must be **continuously monitored and controlled**.

Why: HACCP goes beyond identifying hazards - it establishes the ongoing monitoring, limits, and actions needed to maintain control.

How to use it?

The 7 HACCP principles:

1. Conduct a hazard analysis for each process step (identify hazards & haz. sit.)
2. Determine Critical Control Points (CCPs)
3. Establish appropriate limits for each CCP
4. Establish monitoring procedures for each CCP
5. Establish corrective and preventive actions when limits are exceeded (RCM)
6. Establish verification procedures
7. Establish documentation and record keeping

Example

HACCP for freeze-drying process

CCP #1: Chamber temperature during drying. Critical limit: -25°C to -20°C.

Monitoring: Continuous temperature logging.

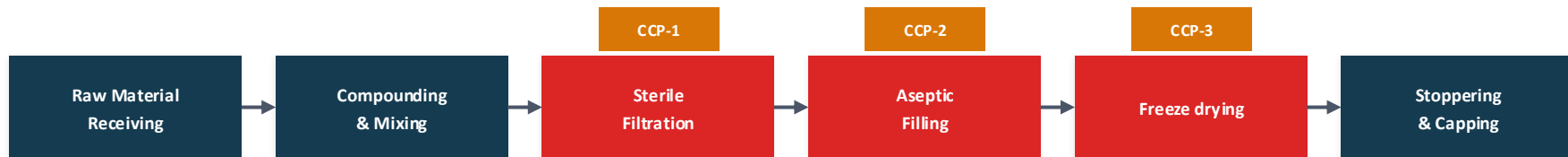
Corrective action: If temperature exceeds limit, quarantine batch and investigate.

Verification: Monthly calibration.

This ensures product safety is built into the process, not just tested at the end.

HACCP Example: Sterile Medical Device Manufacturing

Hazard Analysis & Critical Control Points - identifying CCPs in a manufacturing process and establishing monitoring



CCP-1: Sterile Filtration	CCP-2: Aseptic Filling	CCP-3: Freeze drying
Hazard: Bioburden / sterility	Hazard: Particulate contamination	Hazard: Chemical/physical instability
Critical Limit: 0.22 µm filter integrity test ≥ 3,450 mbar	Critical Limit: ISO 5 environment; < 3,520 particles/m ³ (≥ 0.5µm)	Critical Limit: Shelf temp: -25°C to -20°C primary; chamber pressure < XXX
Monitoring: Pre- and post-use filter integrity test	Monitoring: Continuous particle counter	Monitoring: Continuous temperature sensor + Pressure sensor
Corrective Action: Quarantine batch; re-filter through new validated filter	Corrective Action: Stop fill; investigate source; repeat media fill qualification	Corrective Action: Quarantine batch; stability testing before release
Verification: Annual filter validation; trend of integrity test results	Verification: Semi-annual media fill; environmental monitoring trends	Verification: Monthly calibration; product stability programme

There's at least one more effective risk analysis tool...

(ISO/TR 24971 does not mention it)

Brainstorming with AI

(but keep confidential information to yourselves)

Hands-on exercise

Breakout rooms – Complete the Risk analysis matrix

Phase 2: Hands-on Exercise



20 minutes | Download a copy of the Excel shared in the Teams chat

Complete the risk analysis matrix

Task (in groups of 3-5 persons)

1. For each pre-populated hazard, complete ALL remaining columns in the matrix
2. Write out: Hazard → Sequence of Events → Hazardous Situation → Harm
3. **OPTIONAL (or skip, considering time limit):** Estimate P and S using the scales provided, then determine if the initial risk is acceptable
4. Define risk control measures using the RCM priority order
5. Document verification of BOTH implementation and effectiveness for each control
6. **OPTIONAL (or skip, considering time limit):** Estimate residual risk (P and S) after controls are applied
7. If residual risk is still not acceptable, note the benefit-risk reference
8. Always check: Does your risk control introduce any NEW risks?

Phase 3: Review & Discussion

Let's review your risk analyses. Key questions to consider:

Is the hazardous situation distinct from the harm?

*Common mistake: Conflating these. The situation is the **exposure**; the harm is the **consequence**.*

Is the sequence of events specific and realistic?

Vague sequences like 'device fails' miss the point. Describe the actual chain of events.

Did you consider all 3 tiers of risk control?

Jumping to IFU warnings without documenting why design solutions were infeasible is an audit finding.

Did you verify both implementation AND effectiveness?

*The most common gap - verifying that a control **exists** is not the same as verifying it **works**.*

Did any of your controls introduce new risks?

This is easy to miss. Even minor new risks should be documented and assessed.

Where residual risk is NOT acceptable - did you document the benefit-risk analysis?

For Annex XVI devices, benefit-risk arguments require extra care since the purpose is non-medical.

Key Learnings



Risk analysis is systematic - follow the sequence of events from hazard through to harm



Always consider the risk control hierarchy – Prioritize inherent safety by design or manufacture



Verification of **implementation** and **effectiveness** are two separate required activities



Post-market data continuously feed back into risk management - your risk analysis is a living document



EU 2022/2346 common specifications provide the specific risk catalogue for Annex XVI devices

Questions & Feedback

Part A + B Complete

Please share your feedback on the training: jonas.jagerback@orderlypeople.se

Thank you!

Risk Management & Clinical Investigation

ISO 14155:2020 *Clinical investigation of medical devices for human subjects – Good Clinical Practice*

- **Product risk management**

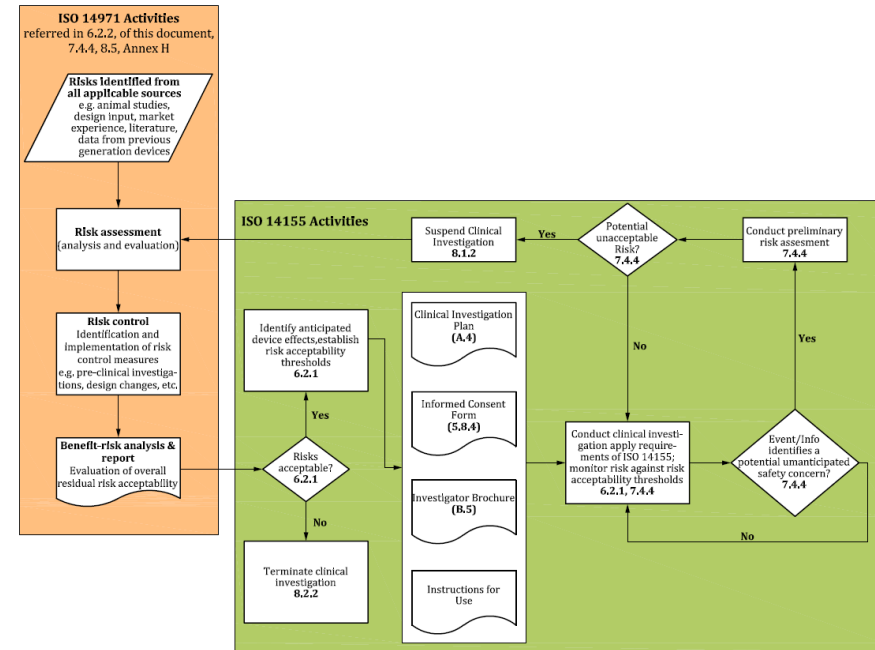
A complete version
Benefit risk analysis established
Input from other than real use

- **Information**

Investigator's brochure
Instructions for use
Clinical investigation plan

- **Clinical investigation process risks**

Other than device risks
Predefine risk acceptability threshold
Plan for risk management during investigation



ISO 14155:2020 - Annex H